

SETTING UP MY FIRST LLM ON WINDOWS

Google Gemma 2B/7B

INTRODUCTION

Setting up an LLM – Google Gemma

To start the installation, I decided to go with Google Gemma3. The specification of the system I used was a Windows 11 Home 64 bits, System Model - Victus by HP Gaming Laptop 15inches

Processor 12th Gen Intel (R), Core i5-12450H (12 CPUs), 2.0GHz and 16GB RAM

Item	Value
OS Name	Microsoft Windows 11 Home
Version	10.0.26100 Build 26100
Other OS Description	Not Available
OS Manufacturer	Microsoft Corporation
System Name	██████████
System Manufacturer	HP
System Model	Victus by HP Gaming Laptop 15-fa1xxx
System Type	x64-based PC
System SKU	9A9N0PA#ABG
Processor	12th Gen Intel(R) Core(TM) i5-12450H, 2000 Mhz, 8 Core(s), 12 Logical Proc...
BIOS Version/Date	AMI F.18, 21/08/2024
SMBIOS Version	3.6
Embedded Controller Version	63.43
BIOS Mode	UEFI
BaseBoard Manufacturer	HP
BaseBoard Product	8C3F
BaseBoard Version	63.43

Fig 1: System information for the proposed System.

What is an LLM?

LLM stands for “**Large Language Model**”. It's a type of artificial intelligence (AI) model designed to understand, generate, and manipulate human-like text based on vast amounts of training data. LLMs are a subset of deep learning models, specifically transformer-based neural networks, and are trained on massive datasets containing books, articles, code, and other text sources.

What is Google's Gemma

Google's Gemma 2B/7B is a lightweight open-weight LLM from Google which is designed to run locally—similar to Meta's Llama 3 or Mistral. It was released in Feb 2024 and is derived from the same technology as Gemini. There are two versions of Gemma.

- Gemma 2B (2 billion parameters) → Good for low-resource devices.
- Gemma 7B (7 billion parameters) → More capable, but still lightweight.

Key Features of Google Gemma:

There are different features of Google Gemma. These features includes: -

- Open weights: Free to use commercially (with terms).
- Local-friendly: They are designed to run on laptops/desktops.
- Performance: They are strong for their size and compete with Llama 2 7B/Mistral 7B.

Setting up Google's Gemma 2B/7B

to setup Google Gemma, we need to setup Ollama.

Ollama is a tool to run large language models (LLMs) locally on your machine (macOS/Linux/WSL). It simplifies downloading, managing, and interacting with open-weight models like:

- Meta's Llama 3 (8B, 70B)
- Google's Gemma (2B, 7B)
- Mistral (7B, 22B)
- Phi-3 (Microsoft's small but powerful model)
- DeepSeek-R1 (a Chinese-English bilingual model)

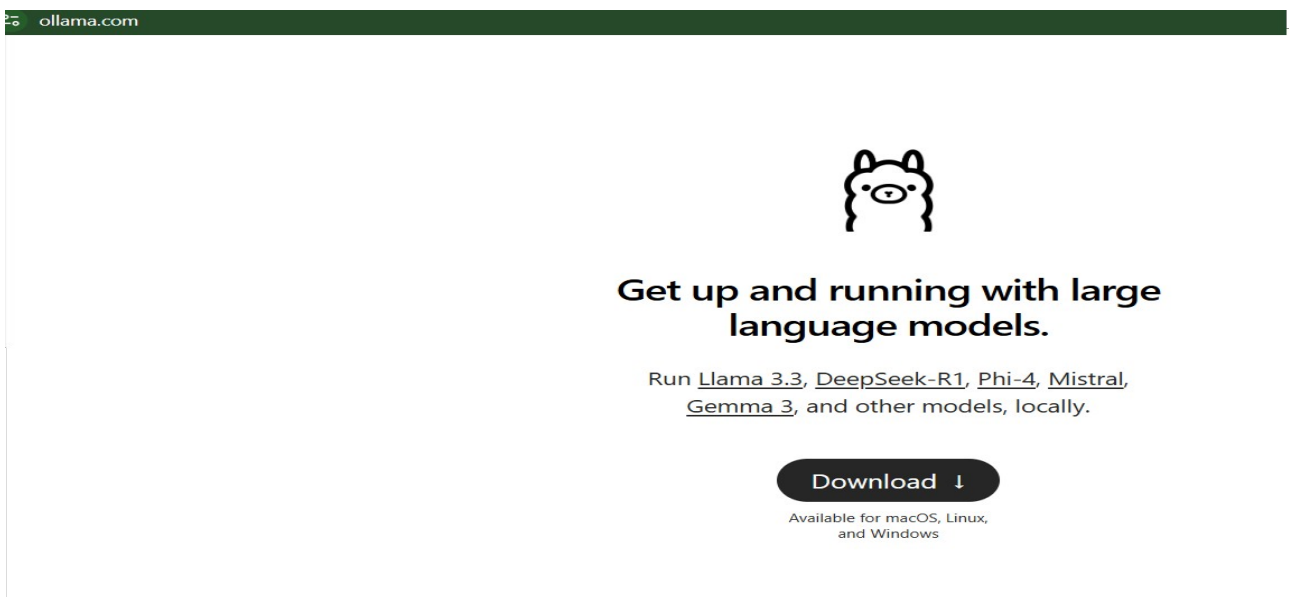


Fig 2: Ollama.com to download our LLM - Gemma3.

To determine if our file has been downloaded, we can use the Command prompt (“CMD”) in verify our version and also check for the installation of “Ollama”

```
Administrator: Command Prompt
Large language model runner

Usage:
ollama [flags]
ollama [command]

Available Commands:
serve      Start ollama
create     Create a model from a Modelfile
show      Show information for a model
run       Run a model
stop      Stop a running model
pull      Pull a model from a registry
push      Push a model to a registry
list      List models
ps        List running models
cp        Copy a model
rm        Remove a model
help      Help about any command

Flags:
-h, --help      help for ollama
-v, --version    Show version information

Use "ollama [command] --help" for more information about a command.

C:\Windows\System32>ollama --version
ollama version is 0.6.2
```

Fig 3: Ollama.com to download our LLM - Gemma3.

To determine the Gemma version to install, we use the model recommended RAM size shown below:-

Setting up and Running our Ollama and Gemma Models

Capacity Requirement					
Summary Table					
Model	Model RAM	System RAM	Buffer	Ideal Total RAM	Recommended RAM
gemma3:1b	~2 GB	2 GB	1 GB	5 GB	8 GB
gemma3:4b	~8 GB	2 GB	2 GB	12 GB	16 GB
gemma3:12b	~16 GB	2 GB	2 GB	20 GB	24-32 GB
gemma3:27b	~24 GB	2 GB	2-4 GB	28-30 GB	32-64 GB

From our table above, we identify that the best Gemma 3 version we intend to use is Gemma 3:4b. This model is multimodal meaning it can support both text and images. We also selected this model because of our RAM size is 16GB. To setup, we go to <https://ollama.com/search>.

We copy our preferred Gemma model and paste in our CMD prompt and run

gemma3

The current, most capable model that runs on a single GPU.

vision 1b 4b 12b 27b

3M Pulls Updated 7 days ago

4b

17 Tags

ollama run gemma3:4b

1b	815MB	a2af6cc3eb7f · 3.3GB
4b	3.3GB	parameters 4.3B · quantization Q4_K_M 3.3GB
12b	8.1GB	nd_of_turn>"], "temperature": 1, "top_k": 64, "t...
27b	17GB	77B
View all		358B
license	Gemma Terms of Use Last modified: February 21, 2024 By using, re... 8.4kB	

Readme

Fig 4: Ollama.com/search to search and select our LLM – Gemma3.

gemma3

The current, most capable model that runs on a single GPU.

vision 1b 4b 12b 27b

3M Pulls Updated 7 days ago

4b

17 Tags

ollama run gemma3:4b

Updated 7 days ago		a2af6cc3eb7f · 3.3GB
model	arch gemma3 · parameters 4.3B · quantization Q4_K_M	3.3GB
params	{ "stop": ["<end_of_turn>"], "temperature": 1, "top_k": 64, "t...	77B
template	{{- range \$i, \$_ := .Messages }} {{- \$last := eq (len (slice \$.M...	358B
license	Gemma Terms of Use Last modified: February 21, 2024 By using, re... 8.4kB	

```
Administrator: Command Prompt
ollama [flags]
ollama [command]

available Commands:
serve      Start ollama
create     Create a model from a Modelfile
show       Show information for a model
run        Run a model
stop       Stop a running model
pull       Pull a model from a registry
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list       List models
ps         List running models
cp         Copy a model
rm         Remove a model
help       Help about any command

flags:
-h, --help      help for ollama
-v, --version   Show version information

use "ollama [command] --help" for more information about a command.

C:\Windows\System32>ollama list
NAME      ID          SIZE      MODIFIED
gemma3:4b a2af6cc3eb7f 3.3 GB    3 hours ago

C:\Windows\System32>ollama run gemma3:4b
```

Fig 5: Using our CLI to setup Gemma 4b (our LLM of our choice).

```
C:\Windows\System32>ollama run gemma3:4b
pulling manifest
pulling aeda25e63ebd... 47% [progress bar] 1.6 GB/3.3 GB 8.6 MB/s 3m27s
```

Fig 6: setting up Gemma 4b (our LLM of our choice).

The setup files are pulled for install on the local machine. However it must be noted that all interactions with the LLM would be at the command line layer.

Running Gemma3 in Command Line

We can run Gemma3 in Command line. To carry out this operation. We identify the gemma3 version using the “list” command. Using the “ollama run gemma3: [version_No]” enables the running of our instance.

```
C:\Users\Kay Freeman>ollama list
NAME      ID          SIZE      MODIFIED
gemma3:4b a2af6cc3eb7f 3.3 GB    3 hours ago

C:\Users\Kay Freeman>ollama run gemma3:4b
>>> Send a message (/? for help)
```

Fig 7: Running Gemma 4b (our LLM of our choice).

Giving our LLM a question in the CMD format returns results as shown below

```
C:\Users\Kay Freeman>ollama run gemma3:4b
>>> how do i run a race?
Okay, let's break down how to run a race, from preparation to crossing that finish line! Here's a comprehensive guide, broken down into categories:

**1. Before the Race - Preparation is Key!**

* **Choose Your Race:**
  * **Distance:** Start with a distance you're comfortable with. 5K (3.1 miles) is a great beginner. Marathons and marathons are much more demanding.
  * **Type:** Consider a road race (on pavement), trail race (over uneven terrain), or fun run (often with a more relaxed pace).
```

Fig 8: Running Gemma 4b (our LLM of our choice).

Alternatively, to create a user interface, we would have to setup an interactive UI which can be gotten from Interactive Web UI (<https://docs.openwebui.com/getting-started/>).

Setting up Open Web UI

To enable and use Open Web UI, we need to setup Docker.

Docker is an open-source platform that allows developers to package, distribute, and run applications in isolated environments called containers. Containers bundle an app with all its dependencies (libraries, system tools, code) so it runs consistently across different computing environments (like your laptop, a cloud server, or a teammate's machine).

Why Use Docker?

- ✓ Consistency – Runs the same way on any machine ("Works on my machine? Now it works everywhere!").
- ✓ Isolation – Apps run in separate containers, avoiding dependency conflicts.
- ✓ Portability – Deploy seamlessly from a laptop to cloud servers (AWS, Azure).
- ✓ Efficiency – Containers start in seconds (vs. minutes for VMs) and use fewer resources.

Setting up Docker

To setup Docker, we download the setup file from:-

<https://www.docker.com/products/docker-desktop/>

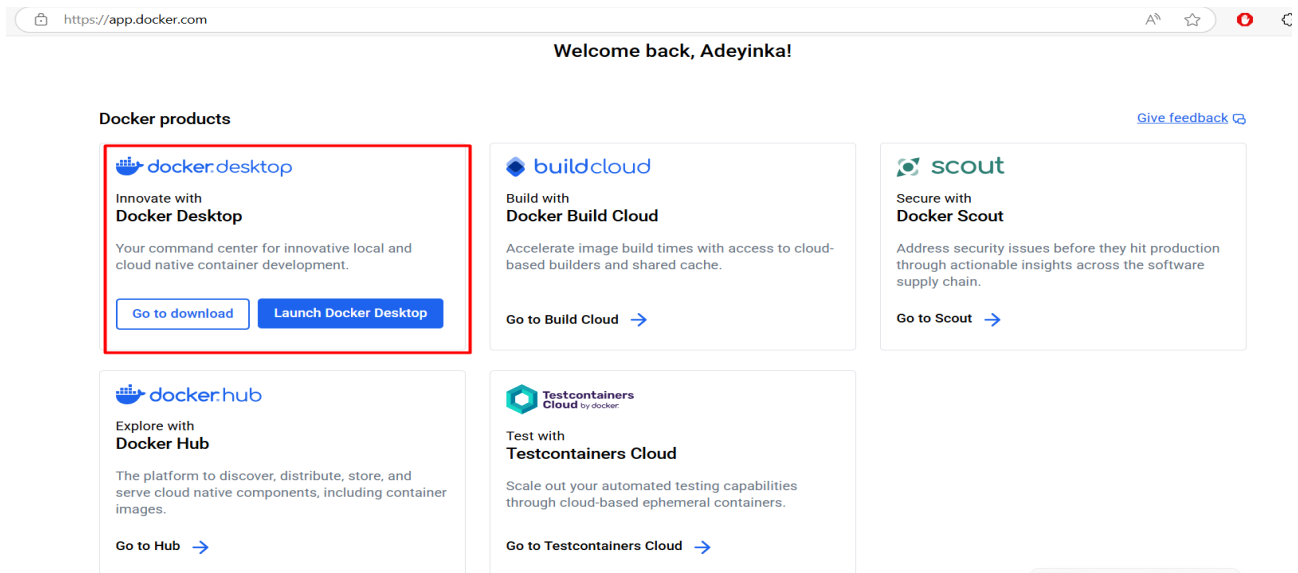


Fig 8: Download Docker for our WebGUI (our LLM of our choice).

we install the Docker Desktop for windows as shown below: -

[Home](#) / [Manuals](#) / [Docker Desktop](#) / [Setup](#) / [Install](#) / Windows

Install Docker Desktop on Windows

Docker Desktop terms

Commercial use of Docker Desktop in larger enterprises (more than 250 employees OR more than \$10 million USD in annual revenue) requires a [paid subscription](#).

This page contains the download URL, information about system requirements, and instructions on how to install Docker Desktop for Windows.

Docker Desktop for Windows - x86_64

Docker Desktop for Windows - Arm (Beta)

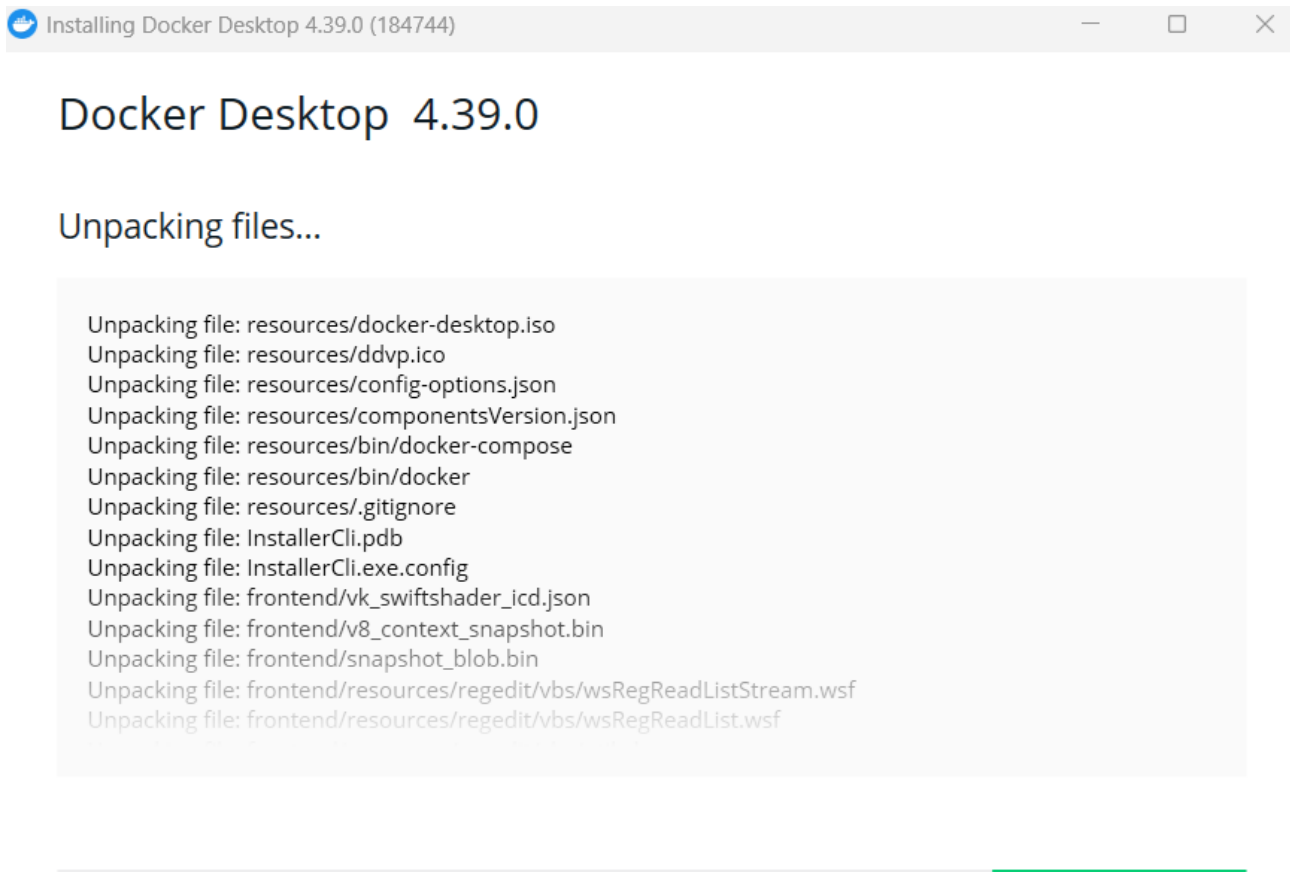


Fig 9: Installing Docker for our WebGUI (our LLM of our choice).

After setting up Docker, We use the Terminal to pull the Open Web GUI using the code displayed below

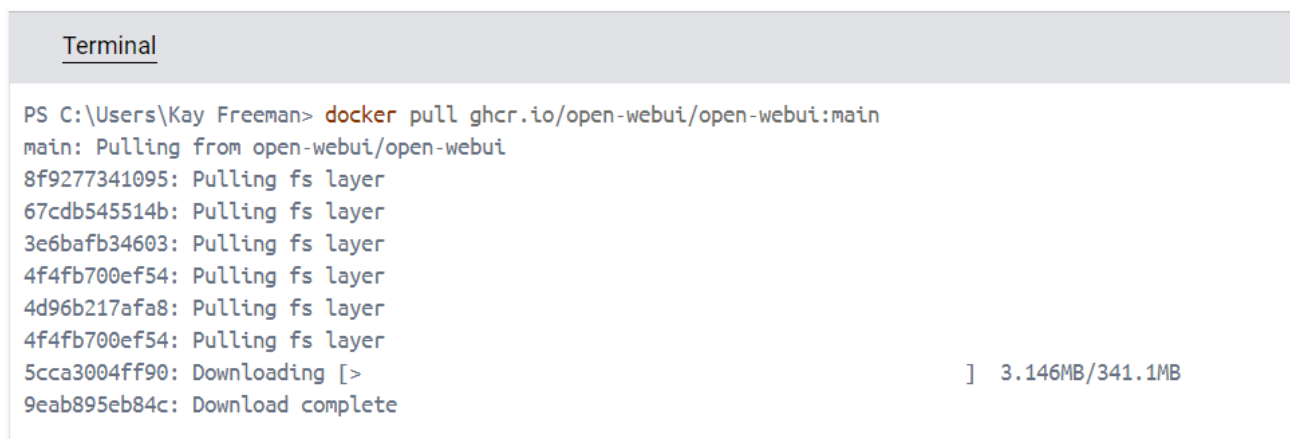
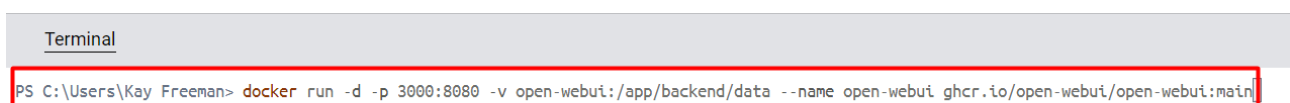


Fig 10: Installing Docker for our WebGUI (our LLM of our choice)....



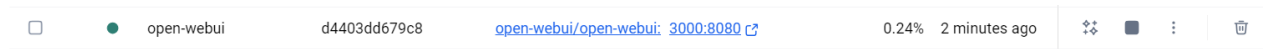


Fig 11: Running Open WebUI on Docker (our LLM of our choice)

Running our LLM (Gemma 4b)

To run our LLM, we run our URL. “<http://localhost:3000/auth?redirect=%2F>” our browser of choice.

A screenshot of a web browser showing the Open WebUI authentication page. The browser's address bar shows 'localhost:3000/auth?redirect=%2F'. The page has a dark background with a nebula image and the text 'Pioneer insights' and '1 person you are'. Below this is a white box with the heading 'Get started with Open WebUI' and a note: 'Open WebUI does not make any external connections, and your data stays securely on your locally hosted server.' There are three input fields: 'Name' (placeholder 'Enter Your Full Name'), 'Email' (placeholder 'Enter Your Email'), and 'Password' (placeholder 'Enter Your Password'). At the bottom is a 'Create Admin Account' button.

localhost:3000/auth?redirect=%2F

New Tab TECH DAYO_CRYPTO Onboarding 100 Great books... Screen for Quali... About - Compoundi... WALLET TRAC

Pioneer insights
1 person you are

Get started with Open WebUI

① Open WebUI does not make any external connections, and your data stays securely on your locally hosted server.

Name
Enter Your Full Name

Email
Enter Your Email

Password
Enter Your Password

Create Admin Account

Fig 13: Running our IP in our preferred browser.

Running our LLM (Gemma 4b) in our preferred browser

Once launched, our Gemma 4b instance works directly on our browser. We can ask questions using text and images.

≡ gemma3:4b ▾ +
Set as default

oi gemma3:4b

How can I help you today?



 Code Interpreter



↗ Suggested

Overcome procrastination

give me tips

Explain options trading

if I'm familiar with buying and selling stocks

Help me study

vocabulary for a college entrance exam